



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ANISH NAIR

BATCH: A1

PRN:25070122037

Subject: Programming with Java

ASSIGNMENT No : 8

TITLE :Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Outcome

```

1  class Vehicle {
2      String vehicleNumber;
3      String model;
4
5      Vehicle(String vehicleNumber, String model) {
6          this.vehicleNumber = vehicleNumber;
7          this.model = model;
8      }
9
10     void displayVehicleInfo() {
11         System.out.println("Vehicle Number: " + vehicleNumber);
12         System.out.println("Model: " + model);
13     }
14 }
15
16 class Insurance extends Vehicle {
17     double premiumAmount;
18
19     Insurance(String vehicleNumber, String model, double premiumAmount) {
20         super(vehicleNumber, model);
21         this.premiumAmount = premiumAmount;
22     }
23
24     void displayInsuranceInfo() {
25         super.displayVehicleInfo();
26         System.out.println("Premium Amount: Rs. " + premiumAmount);
27     }
28 }
29
30 public class VehicleInsuranceDemo {
31     public static void main(String[] args) {
32         Insurance policy = new Insurance("MH12AB1234", "Honda City", 8500.0);
33         policy.displayInsuranceInfo();
34     }
35 }

```

```

Vehicle Number: MH12AB1234
Model: Honda City
Premium Amount: Rs. 8500.0

```

```

1  class Employee {
2      String name;
3      double salary;
4
5      Employee(String name, double salary) {
6          this.name = name;
7          this.salary = salary;
8      }
9
10     void display() {
11         System.out.println("Employee Name: " + name);
12         System.out.println("Salary: Rs. " + salary);
13     }
14 }
15
16 class Manager extends Employee {
17     double bonus;
18
19     Manager(String name, double salary, double bonus) {
20         super(name, salary);
21         this.bonus = bonus;
22     }
23
24     @Override
25     void display() {
26         super.display();
27         System.out.println("Bonus: Rs. " + bonus);
28         System.out.println("Total Pay: Rs. " + (super.salary + bonus));
29     }
30 }
31
32 public class EmployeeManagerDemo {
33     public static void main(String[] args) {
34         Manager mgr = new Manager("Anish Kulkarni", 55000.0, 10000.0);
35         mgr.display();
36     }
37 }

```

```

Employee Name: Anish Kulkarni
Salary: Rs. 55000.0
Bonus: Rs. 10000.0
Total Pay: Rs. 65000.0

```

Conclusion : The super keyword lets Manager and Insurance call their parent class constructors (Employee, Vehicle) and reuse/extend parent methods (display, displayVehicleInfo) without duplicating existing code.

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.
2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.